

11 — OSV 10 kn: the Full Five-Mode Pipeline

Parametric Offshore Support vessel: curve 10 kn → **1 500 kW**, SM 15 % → effective **1 725**. Plant: ME 2×15 000 (Diesel 2-stroke Medium, id 2) · SG 2 000 · AE 4×1 000. Modes: Transit 4 000 h (hotel 220) · DP 2 360 h (thrust 3 500, hotel 300, Calm) · Port 1 200/150 · Anchor 800/200 · Maneuvering 400 (500/250). Battery 1260, Transit only. Total 8 760 h.

Battery Contribution (Transit only)

Propulsion H = $1\,725 \times 5\% = \mathbf{86.25}$ → J $86.25 \times 0.35 = \mathbf{30.2}$, L **56.1** · Hotel H = 4.4 → J 0.22, L 4.18. Tiles: **SR 60.2 / PS 30.4**. (A 1 260 kW battery against a 90.7 kW total swing — saturated 14× over.)

Power Demands (five rows)

Mode	Propulsion'	Hotel (all SG, AE=0!)	ME power	Energy kWh
Transit	$1\,725 + 56.1 = 1\,781$	224.2	2 005.2 (13.37 %)	8 020 970
DP	3 500	300	3 800 (25.33 %)	8 968 000
Port	0	150	150 (1 %)	180 000
Anchor	0	200	200 (1.33 %)	160 000
Maneuvering	500	250	750 (5 %)	300 000

Header: ME $17\,628\,970 / 8\,760 = \mathbf{2\,012}$ · **AE = 0 for all 8 760 h** (SG-forced rule: the ME spins even in port/anchor at ~1 % load; the 4 AEs never run — observation #1 at its clearest).

Per-mode FOC with the actual curve (id 2: 5 %→175, 25 %→174.476, 30 %→172.568 ...)

Mode	SFOC(load)	FOC t/h	× hours = t
Transit	174.78 @13.37 %	0.35048	1 401.9
DP	174.35 @25.33 %	0.66252	1 563.6
Port	175.10* @1 %	0.02627	31.5
Anchor	175.10* @1.33 %	0.03502	28.0
Maneuvering	175.0 @5 %	0.13125	52.5

*extrapolated below the curve's lowest point. **IL1 total = 3 077.5 t/yr ✓**

Baseline, savings, tiers

Transit has only **2 valid combos** (SG covers the hotel → AEs idle → rejected) ⇒ the 3rd-from-worst rule clamps to index 0 = the optimal itself. Baseline **3 081.6** → savings **4.1 t/yr (0.1 %)**, negative ROI — flat SFOC + 2 near-identical rows = nothing to optimize. IL2 adds 0 (single generator carries hotel). **IL3 = 36.3 t/yr** — the only tier with real value: DRC variation $500 - 0.22$ (battery) → $\times 0.8$ → **±400**, 30 cycles/h over 8 760 h.

Battery Benefit

Covered band only 30.4 kW, Transit hours only: **21.2 t/yr = \$16 548**. A heavily oversized battery for this vessel — the model refuses to flatter it.

Takeaway: the whole pipeline end-to-end on a "boring" plant: tiny cascade, clamped baseline, zero L1/L2 value, DRC as the only earner, and the AE fleet parked by the SG-forced rule.